

THE HANS COLER INVENTIONS (PART 2 OF 3)

Translation of a report by Professor W.O. Schumann in Berlin dated April 3, 1926

The apparatus in question principally consists of two parallel spools, which, being bifilarly wound in a special way, are magnetically linked together. One of these spools is composed of copper sheets (the spool is called the plate spool), the other one of a number of this parallel connected isolated wires (spool winding), running parallel at small intervals to the plates. Both spools can be fed by separate batteries. At least two batteries are necessary to put the spools to work.

The spools are arranged in two halves each, according to the bifilar winding system. To the starting points the batteries are attached. To the parallel connected ends the current receivers are attached. Inter-communications are connected between parallel windings of the two halves of the plate spool which contain iron rods with silver connections. These rods are magnetized by a special battery through applied windings (exciter windings).

According to the statement of the inventor, the production of energy takes place principally in these iron rods, and the winding of the spools plays an essential part in it.

As far as it was possible I convinced myself of the conformity of the circuit with the mechanism. The exciter winding is electrically completely separated from the other windings, which was proved when testing the apparatus while being devoid of tension, as well as when testing it while being at work with the aid of an Ohmmeter and a Millivoltmeter.

In order to ascertain possibly concealed sources, the apparatus was searched with a millivoltmeter, without using any external batteries. No effect was perceptible. Besides the apparatus was carried from one room through a corridor to another one in the establishment to exclude the possibility of any secret connection of the mains.

Installed in the apparatus were three current meters for the currents from the three batteries, and furthermore current and volt meters of the soft iron type for the current receivers. Once the two bulbs respectively were employed for this purpose. Besides there was one precision millimeter of the S. & H. turn spool type, also employable as ammeter and voltmeter (Dr. Sp. C.), and a turn spool millivoltmeter of the A.E.G., an ordinary laboratory

instrument, also employable as a voltmeter, (Dr. Sp. A.E.G. C.).

While the apparatus fed two bulbs, the current delivery of the three current-supplying batteries was measured directly at the terminal clamps. The results are as follows:

1. current of the plate battery 48 MA)
2. current of the spool battery 39 MA) Dr. Sp. S. & H. C.
3. current of the exciter battery 170 MA)

The indications of the built in instruments were in unison with the statements of the S.&H. instrument. The total capacity of the batteries of three elements each: $0.257 \times 6 = 1.542$ watt, (the tension of the batteries in reality being below 6 volts). The possible capacity of the two bulbs according the built in soft iron instruments was $3A \times 3.5 v.$, that is 10.5 watts, which means 6.7 fold of the above mentioned capacity.

Thereafter only one bulb was connected, and a second one, exactly of the same type, was regulated with a special accumulator battery to get the same light. All currents were measured with the "Dr. Sp. S. & H. C.", all tensions with the "Dr. Sp. A.E.G. C" and not with the built in instruments.

1. current of the plate battery 28 MA
2. current of the spool circuit battery 23.5 MA
3. current of the exciter battery 180 MA

That means, of reckoned with a 6 volt battery tension, $6 \times 0.232 = 1.392$ watts. The possible capacity of a bulb, fed by a special battery, and showing the same light power was 4.5 volts $\times$ 1.5 amps - 6.75 watts. The proportion of both efficiencies was 4.85.

Both instruments used for testing (measuring) from S. & H. and from the A.E.G. were then compared with each other by a current voltmeter, the highest possible deviation being less than 10% which can, therefore, scarcely influence the result.

The apparatus, according to the statement of the inventor, is adapted now for current increase. The bulb tension of 3-5 volts is less than the tension of the feeding batteries. By changing over in the interior it would also be possible to use it for an increase of tension.

Then there was a test with the aid of the Dr. Sp. millivoltmeter from the A.E.G. to try the decrease of tension in the single winding halves of the plate

pool on the right as well as on the left side of the mechanism. (The form of the spool is that of a long small rectangle.)

**Tensions in Millivolt:
Right Side:**

1. -	2. 0.24
3. -	4. 2.6
5. 8.4	6. 24
7. 22	8. 24.6
9. 26	10. 25

**Winding from the top downward:
Left Side**

1. 0.34	2. -
3. 2.8	4. -
5. 12.4	6. 22
7. 28	8. 100-140 fluctuates much
9. 46	10. 30

The tensions are distributed very unequally, which must be due to current increases and decreases through the iron cores.

Further measurements on the single parts could not be carried on because of the built in parts in the apparatus. Any variations of the regulation resistances were also not undertaken, because the inventor stated that the apparatus was very sensitive in its installation, especially with regard to the magnetic conditions of the iron cores, and that proper treatment would cause interferences which would be very difficult and time consuming to eliminate.

The next day I received a hotwire voltmeter "H. & B." (H. Dr. TH), and further a precision millimeter "S. & H." (Dr. Sp. TH) at the Technical High School of Charlottenburg. With these instruments and those of the previous day the following tests were made:

1. Plate circuit	28 MA	Dr. Sp. T H
Lamp current	1.52 A	Dr. Sp. S. & H. C.
	1.60 A	H Dr. T H
	1.3-1.4	Built in soft iron instrument
Lamp tension	4.05 V	H Dr. T H
Lamp tension	4 V	Built in soft iron instrument
2. Spool circuit	28.5-30 M	Dr. Sp T H
Lamp current	1.47 A	Dr. Sp S. & H. C
	1.56 A	HDTH
Lamp tension	3.8 V	H Dr. T H
3. Exciter circuit	0.173A	Dr. Sp. S. & H. C
Lamp current	1.5 A	H Dr. T H
Lamp tension	3.75-4V	H Dr. T H
Spool circuit	30-30.5 MA	Dr. Sp T H

Once again a comparison of capacity was carried through by using an equal lamp to that in the apparatus accumulator, and bringing it to the same

brightness, as judged by the eye.

Lamp in accumulator circuit:

tension	4 V	Dr. Sp S. & H. C
	3.3 V	Dr. Sp A.E.G. O
current	1.5 A	Dr. Sp S. & H. C

Lamp in apparatus:

tension	3.85-4.0 V	H Dr. T H
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current	1.59	H Dr. T H
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Current in spool circuit	27-28.5 MA	Dr. Sp T H
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Thereafter the instruments with the measured current — and tension — figures were compared with each other:

1. Tensionmeter parallel on two accumulators:	
Dr. Sp A.E.G. C	3.2 V
H Dr. T H	3.8 V
Dr. Sp S. & H. C	4.2-4.3 V
2. Voltaelectrometer in line:	
H Dr. T H	1.47 A
Dr. Sp S. & H. C	1.46-1.47

Even when taking into account the errors of the instruments, the resulting multiplication of energy, in principle, does not undergo any alteration through the apparatus. Judging from the conformity of the hotwire and turnspool instruments in the lamp circuit, it is continuous current one has to deal with. A test of current from the batteries by way of hotwire measurement was not possible.

As a striking fact the spool circuit, having been at first always switched on alone, received a current of 104 MA. As soon as plates and exciter circuit additionally and simultaneously were switched on, as, according to the inventor, the apparatus demands it the current in the spool comes down to about 27 MA.

A definite judgment about the apparatus must be reserved by me until all parts have been singly tested, and until variations in the connections in the load capacity have been undertaken.

After the present examination, carried through as carefully as the limited possibilities of experimentation permitted, I must surmise that we have to face the exploitation of a new source energy whose further developments can be of immense importance. The apparatus was visible and accessible in all of its essential parts. The inventor agreed quite willingly to each trial as long as, according to his statement, no harm would be done to the working of the apparatus.

I do not believe in a deception. I deem it expedient to put the apparatus to a further test, and I believe that a further development of the apparatus, and an assistance given to the inventor, will prove justified and of great importance.